

smMIP Pipeline Overview

smMIP Analysis Pipeline Overview

Quick Start

```
# Basic pipeline (Stages 1-6: FASTQ to mutation calls)
bash /path/to/smMIP-Pipeline/Master_pipeline.sh \
  --directory /path/to/experiment \
  --fastq_dir /path/to/experiment/RAW_FASTQ

# Full pipeline with CHIP analysis (Stages 1-8)
bash /path/to/smMIP-Pipeline/Master_pipeline.sh \
  --directory /path/to/experiment \
  --fastq_dir /path/to/experiment/RAW_FASTQ \
  --chip_analysis
```

Pipeline Stages

Stage	Script	Description
1	BWA.sh	Align paired-end reads to hg19 reference
2	filter_bam.sh	Filter for quality (MAPQ>=50, properly paired)
3	Read_Processing.sh	Map reads to smMIP probes, extract UMIs
4	Level_Bas_Calls.sh	Generate variant pileups at smMIP level
5	Annotate_SNVs.R	Annotate panel with gene/protein/COSMIC info
6	calling_mutations.R	Call mutations using statistical error modeling
7	CHIP_analysis.R	Population-level CHIP analysis (oncoplots, VAF heatmaps)
8	generate_patient_reports.R	Generate individual patient HTML/PDF reports

Command-Line Parameters

Required Parameters

Parameter	Description
<code>--directory</code>	Base experiment directory (outputs go here)
<code>--fastq_dir</code>	Directory containing raw FASTQ files

Optional Parameters

Parameter	Default	Description
<code>--panel</code>	Universal/ Myeloid_Panel_Targets.chr.txt	Panel design file
<code>--ref</code>	Universal/hg19/hg19.fa	Reference genome
<code>--annotated_panel</code>	Universal/ annotated_Myeloid_Panel_Targets.txt	Pre-annotated panel
<code>--config</code>	{directory}/config.txt	Sample configuration file
<code>--threads</code>	6	Threads per sample
<code>--parallel_jobs</code>	4	Samples to run in parallel
<code>--use_parallel</code>	false	Use parallel versions of scripts (flag)
<code>--force</code>	false	Force re-run all stages even if outputs exist
<code>--start</code>	1	Start at this stage (1-8)
<code>--stop</code>	6	Stop after this stage (1-8)

CHIP Analysis Parameters (Stages 7-8)

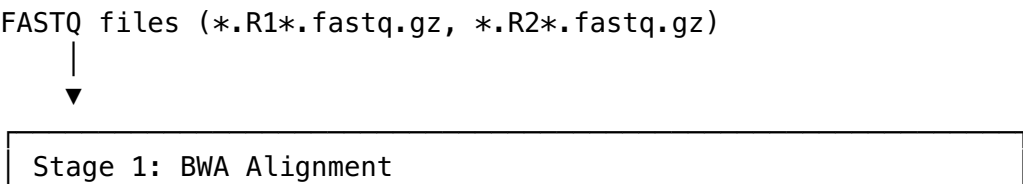
Parameter	Default	Description
<code>--chip_analysis</code>	-	Enable CHIP analysis (sets <code>--stop</code> 8)
<code>--chip_vaf</code>	0.01	VAF threshold for CHIP filtering
<code>--chip_pval</code>	0.05	P-value threshold for CHIP filtering

Parameter	Default	Description
--skip_reports	false	Skip patient report generation (Stage 8)
--skip_clinvar	false	Skip ClinVar API queries (faster reports)
--report_format	html	Report format: html or pdf

Directory Structure

```
{--directory}/
├── bwa_out/                                # Stage 1 output
│   ├── *.bam
│   ├── *.bam.bai
│   ├── *.flagstat.txt
│   └── _logs/
├── filtered_bam/                          # Stage 2 output
│   ├── *.filtered.bam
│   ├── *.filtered.bam.bai
│   └── _logs/
├── Read_Processing/                      # Stage 3 output
│   ├── {sample}/
│   │   ├── {sample}_clean.bam
│   │   ├── {sample}_filtered_read_counts.txt
│   │   ├── {sample}_raw_coverage_per_smMIP.txt
│   │   └── {sample}_UMI_usage_per_smMIP.txt
│   ├── pileup/                          # Stage 4 output
│   │   ├── {sample}_raw_pileup.txt
│   │   └── {sample}_sscs_pileup.txt
│   └── _logs/
├── results/                              # Stage 6 output
│   └── called_mutations.txt
├── CHIP_analysis/                        # Stages 7-8 output
│   ├── CHIP_oncoplot.pdf/png
│   ├── VAF_heatmap.pdf/png
│   ├── VAF_dotplot.pdf/png
│   ├── co_mutation_heatmap.pdf
│   ├── gene_mutation_summary.csv
│   ├── variant_annotation_summary.csv
│   ├── co_mutation_fisher_results.csv
│   └── patient_reports/
│       ├── {sample}_CHIP_Report.html
│       └── annotation_cache/
└── config.txt                            # Sample configuration
```

Pipeline Flow Diagram



Script: BWA.sh / BWA_parallel.sh
Input: *R1*.fastq.gz, *R2*.fastq.gz
Output: *.bam, *.bam.bai, *.flagstat.txt



Stage 2: BAM Filtering
Script: filter_bam.sh / filter_bam_parallel.sh
Input: *.bam from Stage 1
Output: *.filtered.bam
Filter: MAPQ >= 50, properly paired (-f 2)



Stage 3: Read Processing / UMI Extraction
Script: Read_Processing.sh / Read_Processing_parallel.sh
R Script: map_smMIPs_extract_UMIs.R
Input: *.filtered.bam + Panel_Targets.txt
Output: Per sample directory containing:

- {sample}_clean.bam
- {sample}_filtered_read_counts.txt
- {sample}_raw_coverage_per_smMIP.txt
- {sample}_UMI_usage_per_smMIP.txt



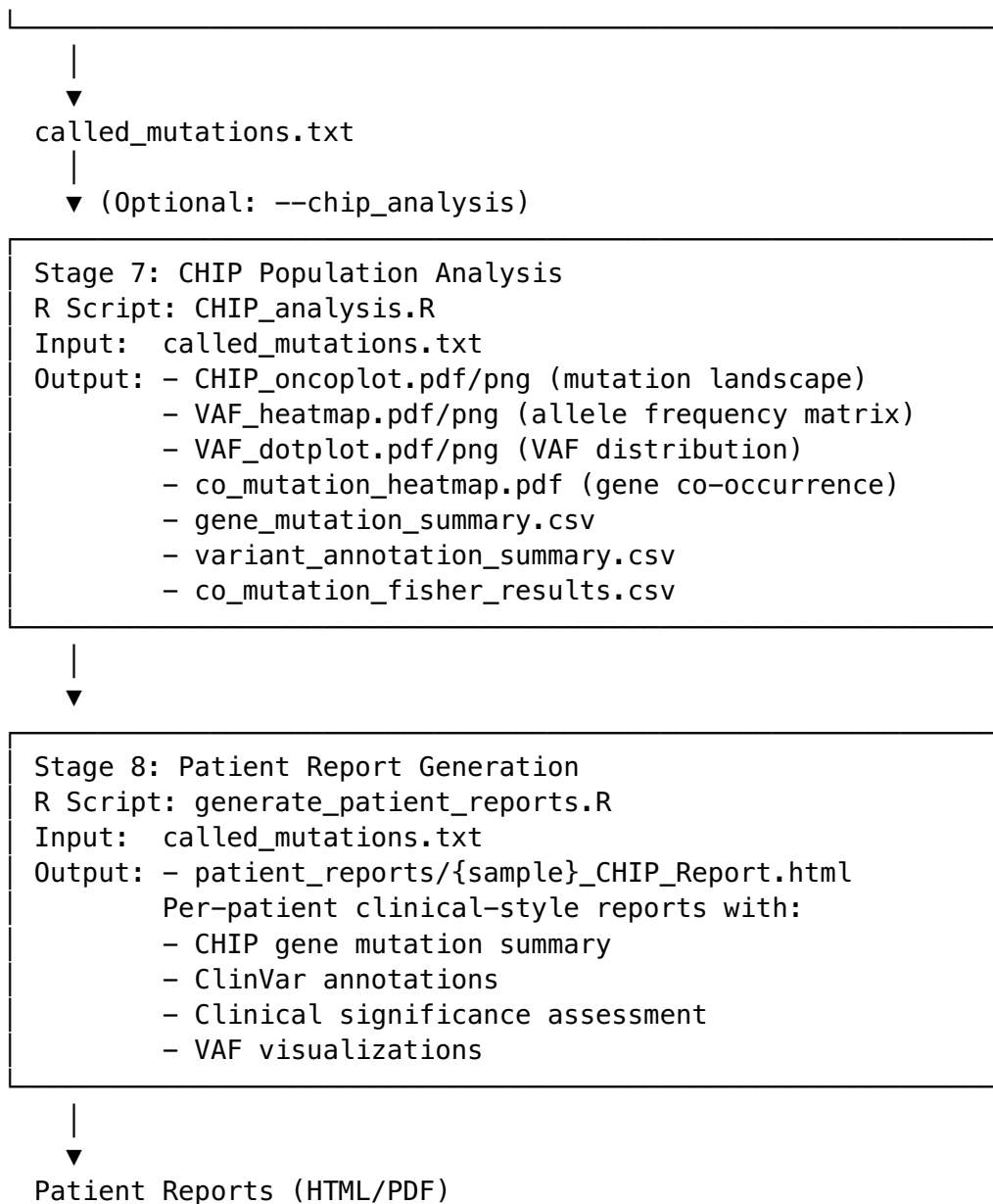
Stage 4: Pileup Generation (Level Base Calls)
Script: Level_Bas_Calls.sh / Level_Bas_Calls_parallel.sh
R Script: smMIP_level_raw_and_consensus_pileups.R
Input: {sample}_clean.bam + Panel_Targets.txt
Output: - {sample}_raw_pileup.txt
 - {sample}_sscs_pileup.txt (consensus)



Stage 5: Panel Annotation (ONE-TIME per panel)
R Script: Annotate_SNVs.R
Input: Panel_Targets.txt
Output: annotated_Panel_Targets.txt
Annotations: gene, protein, COSMIC, MAF, CADD scores



Stage 6: Mutation Calling
R Script: calling_mutations.R
Input: - Pileup folder (*_raw_pileup.txt, *_sscs_pileup)
 - Configuration file (sample info)
 - Annotated panel file
Output: called_mutations.txt



Configuration File Format

The configuration file (config.txt) describes sample metadata for mutation calling.

Format

Tab-delimited file with columns:

Column	Description
id	Sample ID (must match pileup filenames)
type	Sample type: case or control
replicate	Technical replicate pairing (or NA)

Example

```
id type replicate
Sample1 case NA
Sample2 case NA
Sample3 control NA
Sample4 case rep1
Sample5 case rep1
```

Auto-Generation

If `--config` is not provided, the pipeline will: 1. Scan the pileup directory for `*_raw_pileup.txt` files 2. Extract sample IDs from filenames 3. Generate a `template config.txt` with all samples as case 4. **Stop and prompt user to edit the file** 5. Re-run with `--start 6` to continue

Mutation Calling Parameters

Key parameters for `calling_mutations.R`:

Parameter	Default	Description
<code>-s</code>	Required	Path to pileup folder
<code>-f</code>	Required	Configuration file (sample layout)
<code>-a</code>	Required	Annotated panel file
<code>-b</code>	“sum”	Binomial error rate method (“sum” or “max”)
<code>-g</code>	10	Min coverage for overlap (R1/R2)
<code>-m</code>	0.001	MAF cutoff to exclude known SNPs
<code>-v</code>	0.05	VAF cutoff for error modeling
<code>-p</code>	0.05	P-value cutoff (Bonferroni corrected)
<code>-d</code>	2	Fold-difference for cross-sample VAF
<code>-t</code>	1	Number of threads

Output: `called_mutations.txt`

Key columns in the final mutation calls:

Column	Description
<code>sample_ID</code>	Sample identifier
<code>smMIP</code>	smMIP probe name
<code>chr</code>	Chromosome
<code>pos</code>	Position (hg19)
<code>ref</code>	Reference allele
<code>alt</code>	Alternative allele
<code>gene</code>	Gene symbol

Column	Description
protein	Protein change (e.g., p.V600E)
cosmic	COSMIC ID if known
maf	Minor allele frequency (population)
variant_type	SNV, insertion, deletion
cadd_scaled	CADD pathogenicity score
P-value	Raw p-value
P-value.Bonferroni	Bonferroni-corrected p-value
non.ref.counts	Non-reference allele counts
total.depth	Total read depth
allele.frequency	Variant allele frequency
SSCS.*	Consensus-based metrics
flags	Quality flags

Output: CHIP Analysis (Stages 7-8)

Stage 7 Outputs

File	Description
CHIP_oncoplot.pdf/png	Mutation landscape visualization showing genes x samples
VAF_heatmap.pdf/png	Heatmap of variant allele frequencies
VAF_dotplot.pdf/png	Distribution of VAFs per gene
co_mutation_heatmap.pdf	Gene co-occurrence/mutual exclusivity analysis
gene_mutation_summary.csv	Frequency statistics per gene
variant_annotation_summary.csv	Per-variant details with protein changes
co_mutation_fisher_results.csv	Fisher's exact test results for co-mutation

Stage 8 Outputs

File	Description
patient_reports/{sample}_CHIP_Report.html	Individual patient reports with: - Clinically significant mutations (VAF $\geq$ 1%) - Emerging clones (VAF < 1%) - CHIP gene clinical summaries - ClinVar pathogenicity annotations - VAF visualizations

CHIP Analysis Parameters

Parameter	CLI Flag	Default	Description
VAF threshold	--chip_vaf	0.01	Minimum VAF for CHIP filtering
P-value threshold	--chip_pval	0.05	Maximum p-value for filtering
Skip ClinVar	--skip_clinvar	false	Use local knowledge base only
Report format	--report_format	html	Output format: html or pdf

Usage Examples

Full Pipeline Run

```
bash Master_pipeline.sh \  
  --directory /path/to/experiment \  
  --fastq_dir /path/to/fastqs
```

Parallel Processing (Recommended for Large Datasets)

```
bash Master_pipeline.sh \  
  --directory /path/to/experiment \  
  --fastq_dir /path/to/fastqs \  
  --use_parallel \  
  --parallel_jobs 4 \  
  --threads 6
```

Resume from Specific Stage

```
# After editing config.txt, run only mutation calling  
bash Master_pipeline.sh \  
  --directory /path/to/experiment \  
  --fastq_dir /path/to/fastqs \  
  --start 6 --stop 6
```

Force Re-run All Stages

```
bash Master_pipeline.sh \  
  --directory /path/to/experiment \  
  --fastq_dir /path/to/fastqs \  
  --force
```


Run CHIP Analysis on Existing Mutations

```
# Run only CHIP stages (7-8) on existing called_mutations.txt
bash Master_pipeline.sh \
  --directory /path/to/experiment \
  --start 7 --stop 8
```

Full Pipeline with CHIP Analysis

```
bash Master_pipeline.sh \
  --directory /path/to/experiment \
  --fastq_dir /path/to/fastqs \
  --chip_analysis \
  --report_format html
```

CHIP Analysis with Custom Thresholds

```
bash Master_pipeline.sh \
  --directory /path/to/experiment \
  --start 7 --stop 8 \
  --chip_vaf 0.02 \
  --chip_pval 0.01 \
  --skip_clinvar # Faster, uses local knowledge base only
```

Custom Panel and Reference

```
bash Master_pipeline.sh \
  --directory /path/to/experiment \
  --fastq_dir /path/to/fastqs \
  --panel /path/to/custom_panel.txt \
  --ref /path/to/hg38.fa \
  --annotated_panel /path/to/annotated_panel.txt
```

Resource Requirements

Memory

Stage	Memory per Sample	Notes
1 (BWA)	~8 GB	Scales with reference genome size
2 (Filter)	~4 GB	I/O bound
3 (ReadProc)	~30 GB	R memory-intensive
4 (Pileup)	~25 GB	R memory-intensive
5 (Annotate)	~8 GB	Network-bound (API calls)
6 (Mutations)	~16 GB	Depends on cohort size
7 (CHIP)	~16 GB	Scales with cohort size

Stage	Memory per Sample	Notes
8 (Reports)	~8 GB	Per-sample, ClinVar API calls

Recommended Settings

System RAM	<code>--parallel_jobs</code>	<code>--threads</code>
32 GB	2	4
64 GB	3	6
128 GB	4	6
256 GB	8	6

Dependencies

Command-Line Tools

- bwa (or bwa-mem2)
- samtools
- GNU parallel (for `--use_parallel`)

R Packages

Core Pipeline: - optparse - data.table - parallel - cellbaseR (for annotation) - IRanges - Rsamtools

CHIP Analysis (Stages 7-8): - tidyverse - ComplexHeatmap (Bioconductor) - circlize - RColorBrewer - rmarkdown - knitr - kableExtra - rentrez (for ClinVar API)

Troubleshooting

Out of Memory

- Reduce `--parallel_jobs`
- Ensure `R_MAX_VSIZE` is set (automatically handled by pipeline)

Missing R1/R2 Pairs

- Check FASTQ naming convention matches `*R1*.fastq.gz` pattern
- Use `--r1_glob` and `--r2_token` to customize pattern

Stage Skip Not Working

- Use `--force` to override skip logic
- Check that expected output files exist and are non-empty

Config File Issues

- Ensure tab-delimited format
- Sample IDs must match pileup filenames exactly
- All samples need a type assignment (case or control)